

Solution Design Document (SDD)

Family Banking Program – Saudi Investment Bank (SAIB)

1. Document Control

Item	Details
Project Name	Family Banking Program
Organization	Saudi Investment Bank (SAIB)
Document Type	Solution Design Document (SDD)
Version	1.0
Prepared By	Solution Architecture Team
Source Reference	Business Requirements Document (BRD)
Status	Draft

2. Purpose

This Solution Design Document defines the technical and functional solution design for the SAIB Family Banking Program. The solution enables customers to create and manage family relationships between banking customers and extend banking benefits, segmentation, and lifecycle privileges across linked family members.

The solution covers:

- Family initiation and onboarding
 - Dependent validation via national systems
 - Family linking and de-linking
 - Lifecycle management
 - Segment inheritance
 - Notifications and communication
 - RM/CRM operational journeys
 - Customer self-service journeys
 - Audit and compliance requirements
-

3. Business Objectives

The primary business objectives are:

1. Enable a primary customer to create a family structure.
 2. Allow linking of eligible family members.
 3. Share family banking benefits and segmentation.
 4. Support customer and RM initiated journeys.
 5. Support lifecycle handling such as upgrades, downgrades, and de-linking.
 6. Provide compliance-controlled onboarding and consent flows.
 7. Improve customer retention and household wallet share.
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4. Scope

4.1 In Scope

Segments

- Private Banking
- Platinum
- Gold

Channels

- CRM
- New RIB
- Call Center
- Branch
- Desktop RIB

Processes

- Initiation of linking
 - Validation of dependents
 - Family group creation
 - Viewing family structure
 - De-linking dependents
 - Lifecycle management
 - Notifications
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4.2 Out of Scope

- Lending products

- Payment products
 - Investment products
 - ATM channels
 - New account opening
 - Inter-RM family handling
 - Welcome kits / physical communication
 - Segment physical benefit delivery
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5. High Level Architecture

5.1 Core Systems

System	Responsibility
CRM	RM operational journeys and approval workflows
RIB	Customer self-service journeys
Family Banking Service	Core orchestration and family management
Abshar / DPI Integration	National family validation
CIF / Customer System	Customer master information
Notification Service	SMS, Email, Push notifications
Segmentation Engine	Segment inheritance and recalculation
Audit Service	Activity tracking and compliance logging

5.2 Architecture Style

The solution will follow a service-oriented architecture with API-driven integration.

Architectural Principles

- Stateless APIs
 - Centralized orchestration
 - Event-driven notifications
 - Reusable integration services
 - Audit-first design
 - Secure customer consent handling
 - Arabic and English support
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6. Solution Components

6.1 Family Banking Service

The Family Banking Service is the core orchestration component responsible for:

- Family creation
- Family member linking
- Validation orchestration
- Segment propagation
- Lifecycle event handling
- De-linking operations
- Relationship storage
- Eligibility checks
- Audit tracking

Core Functionalities

Capability	Description
Family Group Creation	Create family structure
Relationship Management	Maintain parent-dependent relationships
Eligibility Validation	Validate members against Abshar
Segment Inheritance	Apply segment changes
De-linking	Remove linked members
Notifications	Trigger communication events
Audit Logging	Capture all activities

6.2 CRM Module

CRM will support RM initiated onboarding.

Features

- Search customer using National ID / Iqama
- Fetch customer information
- Trigger OTP
- Submit family linking requests
- View family structures
- Initiate de-linking
- Receive approval/rejection statuses

- Audit operations

User Roles

Role	Access
RM	Maker
Segmentation Unit	Checker

6.3 Retail Internet Banking (RIB)

RIB will support customer initiated onboarding.

Features

- Customer self-service linking
- OTP verification
- Dependent selection
- Family name management
- View family structure
- De-linking request
- Consent management
- Notification viewing

6.4 Integration Layer

External Integrations

External System	Purpose
Abshar	Family relationship validation
National ID Services	Identity validation
OTP Gateway	Customer verification
Notification Gateway	SMS/Email/Push delivery

7. Data Model

7.1 Family Group Entity

Attribute	Description
Family Group ID	Unique family identifier
Family Name	Customer-defined family name
Primary Customer CIF	Main customer
Segment	Assigned segment
Status	Active / Pending / Closed
Created Date	Creation timestamp
Updated Date	Last modified timestamp

7.2 Family Member Entity

Attribute	Description
Member CIF	Customer CIF
National ID	Customer identifier
Relationship Type	Spouse / Minor / Major
Linked Status	Linked / Delinked
Segment	Current inherited segment
OTP Status	Pending / Verified
Consent Status	Accepted / Rejected

7.3 Audit Entity

Attribute	Description
Audit ID	Unique audit record
Event Type	Create / Update / Delink
User ID	RM or Customer

Attribute	Description
Channel	CRM / RIB
Timestamp	Action time
Remarks	Additional details

8. Functional Design

8.1 Family Linking Journey – RM Initiated

Process Flow

1. RM initiates family request.
2. RM enters National ID / Iqama.
3. System fetches customer profile.
4. RM adds dependent IDs.
5. OTP triggered to primary customer.
6. Customer validates OTP.
7. Family validation performed via Abshar.
8. Eligible dependents identified.
9. Family structure created.
10. Segment inheritance applied.
11. Notifications sent.
12. Audit logs captured.

Validation Rules

Rule	Description
Maximum Dependents	10
Family Group Limit	One family group per customer
OTP Mandatory	Yes
Abshar Validation	Mandatory
Minor Acceptance	Not required
Major Acceptance	Mandatory

8.2 Family Linking Journey – Customer Initiated

Process Flow

1. Customer logs into RIB.
 2. Customer initiates family linking.
 3. Customer enters dependent IDs.
 4. OTP verification initiated.
 5. Validation via Abshar.
 6. Eligible members identified.
 7. Family group created.
 8. Notifications triggered.
 9. Audit logs generated.
-

8.3 Validation Logic

Matching Validation

System Actions

- Validate relationship through Abshar.
- Fetch dependent account details.
- Verify customer eligibility.
- Ensure member not already linked.
- Verify segment eligibility.

Matching Result

Result	Action
Matched	Proceed with linking
Unmatched	Reject request and notify
Service Failure	Retry based on retry policy

8.4 Segment Inheritance Logic

Rules

1. Linked dependents inherit primary customer segment.
2. Higher segment dependents retain higher segment.

3. Upgrades propagate automatically.
 4. Downgrades propagate unless exception rule applies.
 5. Lifecycle changes trigger notifications.
-

8.5 De-linking Process

Customer Initiated

Flow

1. Customer selects dependent.
 2. Confirmation popup displayed.
 3. OTP verification triggered.
 4. De-link request processed.
 5. Segment recalculation executed.
 6. Notifications sent.
 7. Audit captured.
-

RM Initiated

Flow

1. RM searches family structure.
 2. RM selects dependent.
 3. System validates authorization.
 4. De-link operation executed.
 5. Notification sent to all stakeholders.
-

De-linking Rules

Rule	Description
Self Delink	Dependents can delink themselves
OTP Required	Yes
Segment Recalculation	Mandatory
Notification	Mandatory
Audit Logging	Mandatory

8.6 Family Structure View

The solution shall support viewing of family hierarchy.

Viewable Information

Information	Visibility
Family Group Name	Visible
Primary Customer	Visible
Dependent Names	Visible
Last 4 digits Account Number	Visible
Full Account Number	Hidden
Balances	Hidden
Transactions	Hidden

9. UI Design Requirements

9.1 CRM UI

Screens

- Initiate Family Linking
- Validation Screen
- Family Structure View
- De-link Screen
- Family Group Update

Mandatory Fields

Field	Mandatory
National ID / Iqama	Yes
CIF Number	Yes
Dependent ID	Yes
OTP	Yes

9.2 RIB UI

Screens

- Initiate Family Linking
 - Dependent Validation
 - Family Overview
 - De-link Management
 - Notification Center
-

10. Notification Design

10.1 Notification Channels

Channel	Supported
SMS	Yes
Email	Yes
Push Notification	Yes
In-App Notification	Yes

10.2 Notification Events

Event	Trigger
Successful Linking	Family created
Failed Validation	Abshar mismatch
Segment Upgrade	Segment changed
Segment Downgrade	Segment changed
Delinking	Member removed
Major Age Attainment	Minor becomes major

10.3 Notification Templates

Templates must support:

- Arabic
 - English
 - Dynamic placeholders
 - Customer personalization
 - Segment information
 - Audit traceability
-

11. Security Design

11.1 Authentication

- OTP verification mandatory
 - Session validation required
 - Channel authentication enforced
-

11.2 Authorization

User Type	Access
Primary Customer	Full family management
Dependent	Limited self-service
RM	Operational management
Segmentation Unit	Approval authority

11.3 Data Security

- Encrypt sensitive data in transit
 - Encrypt sensitive data at rest
 - Mask account numbers
 - Maintain audit logs
 - Secure API communication
-

12. Compliance Requirements

Mandatory Controls

1. OTP consent required.
 2. Customer acceptance mandatory.
 3. Audit trail mandatory.
 4. De-linking notifications mandatory.
 5. Regulatory logging mandatory.
 6. Consent retention required.
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13. Non-Functional Requirements

13.1 Performance

Requirement	Target
End of Day Processing	Supported
Concurrent Users	Up to 10,000
Monthly Transactions	Up to 1 million

13.2 Availability

System	Availability
RIB	24x7
CRM	Business Hours
Equations	Business Hours

13.3 Scalability

The platform must support:

- Horizontal scaling
- High transaction volume
- Future product onboarding
- Expansion of family banking services

13.4 Reliability

Target system availability:

- 99.9%
-

13.5 Audit

The solution must:

- Log all customer actions
 - Log RM activities
 - Capture notifications
 - Retain compliance evidence
 - Support reporting extraction
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14. Error Handling

Retry Strategy

Scenario	Action
Abshar Timeout	Retry 3 times
OTP Failure	Re-initiate OTP
Notification Failure	Retry asynchronously
API Failure	Circuit breaker handling

Validation Errors

Error	Description
Duplicate Member	Already linked
Invalid Relationship	Not verified by Abshar
Segment Restriction	Customer not eligible
OTP Failure	Authentication failed

15. Reporting Requirements

Reports

Report	Purpose
Family Structure Report	Relationship visibility
Segment Report	Segment inheritance tracking
Audit Report	Compliance audit
Notification Report	Communication tracking

16. API Design

16.1 Family APIs

API	Method	Description
/family/create	POST	Create family structure
/family/validate	POST	Validate members
/family/link	POST	Link dependents
/family/delink	POST	Remove dependents
/family/view	GET	View structure
/family/update	PUT	Update family details

16.2 Integration APIs

API	Description
Abshar Validation API	Relationship verification
OTP API	Customer authentication
Notification API	SMS/Email delivery
Customer Profile API	CIF information

17. Database Design Considerations

Key Tables

- FAMILY_GROUP
 - FAMILY_MEMBER
 - FAMILY_AUDIT
 - FAMILY_NOTIFICATION
 - FAMILY_SEGMENT_HISTORY
 - OTP_TRACKING
-

18. Deployment Design

Environments

Environment	Purpose
DEV	Development
SIT	System Integration Testing
UAT	User Acceptance Testing
PROD	Production

Deployment Strategy

- CI/CD deployment
 - API versioning
 - Backward compatibility
 - Blue/Green deployment preferred
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19. Additional BRD Open Items / Clarifications

The final BRD page contains additional open items that must be tracked in the SDD because they affect compliance, validation, and implementation rules.

19.1 Family Member Opt-Out

Family members must have an option to opt out of the family banking program where required by regulation or internal policy.

Design Impact

- Provide an opt-out action for eligible linked family members.
 - Capture opt-out request channel, timestamp, customer identifier, and reason if applicable.
 - Trigger segment recalculation after opt-out.
 - Notify the primary customer and opted-out member.
 - Maintain audit evidence for compliance review.
-

19.2 Maximum Number of Dependents Clarification

The BRD requires compliance confirmation on the maximum number of dependents that may be added under a single customer. The current working assumption is 10 dependents per primary customer, pending final policy confirmation.

Design Impact

- Configure maximum dependents as a parameter, not hard-coded logic.
 - Default value: 10 dependents.
 - Allow future change through configuration without code deployment.
 - Block additional dependents once the configured limit is reached.
 - Show a clear error message to RM/customer when the limit is exceeded.
-

19.3 Customer Lookup Key for RM Initiation

For RM-initiated family banking requests, the system must provide a lookup key to find the customer before proceeding. Supported lookup keys should include:

- CIF
- National ID
- Iqama ID

Design Impact

- CRM search screen must allow customer lookup using CIF, National ID, or Iqama.
 - Search results must return only eligible customer records.
 - Duplicate or ambiguous search results must require RM selection before proceeding.
 - Customer identity must be validated before family linking is initiated.
 - All search activity must be logged for audit.
-

20. Assumptions

1. Abshar integration APIs are available.
2. OTP service already exists.
3. CRM supports maker-checker workflow.
4. Customer CIF information available centrally.
5. Notification platform supports multilingual templates.

20. Risks

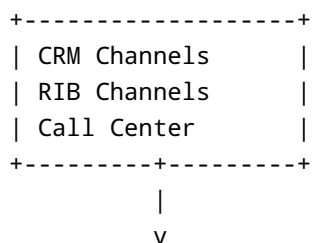
Risk	Impact
Abshar service unavailability	Validation failures
OTP delays	Customer onboarding delay
Segment synchronization issue	Incorrect benefits
Duplicate family structures	Compliance issue

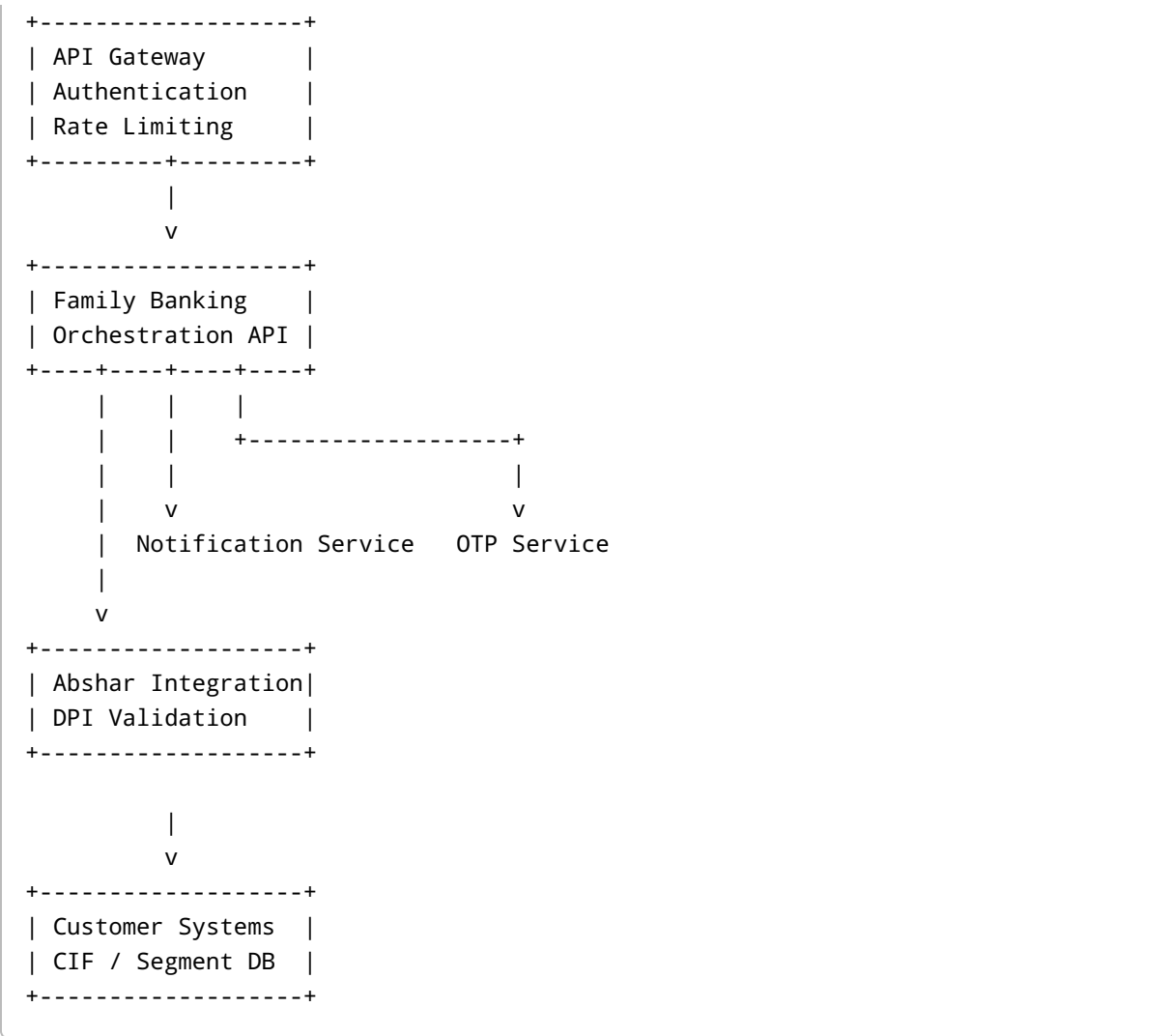
21. Future Enhancements

- Shared family wallets
- Family savings products
- Loyalty pooling
- Joint family investment products
- Advanced analytics dashboards
- AI-based relationship validation

22. Detailed Solution Architecture

22.1 Logical Architecture





22.2 Component Responsibilities

Component	Responsibility
API Gateway	Authentication, throttling, routing
Family Banking API	Main orchestration logic
Validation Engine	Relationship validation
Segment Engine	Segment inheritance
Notification Service	SMS, Email, Push
Audit Service	Audit tracking
Batch Processor	EOD propagation

Component	Responsibility
CRM Adapter	CRM integration
RIB Adapter	Digital channel integration

23. Sequence Diagrams

23.1 RM Initiated Family Linking

```
RM -> CRM
CRM -> Family API
Family API -> Customer System
Customer System -> Family API
Family API -> OTP Service
OTP Service -> Customer
Customer -> CRM
CRM -> Family API
Family API -> Abshar
Abshar -> Family API
Family API -> Segment Engine
Family API -> Notification Service
Family API -> Audit Service
Family API -> CRM
```

23.2 Customer Initiated Linking

```
Customer -> RIB
RIB -> Family API
Family API -> OTP Service
OTP Service -> Customer
Customer -> RIB
RIB -> Family API
Family API -> Abshar
Family API -> Segment Engine
Family API -> Notification Service
Family API -> Audit Service
Family API -> RIB
```

23.3 De-linking Sequence

```
Customer/RM -> Channel  
Channel -> Family API  
Family API -> OTP Service  
Family API -> Segment Engine  
Family API -> Notification Service  
Family API -> Audit Service  
Family API -> Database
```

24. API Specifications

24.1 Create Family API

Endpoint

```
POST /api/v1/family/create
```

Request

```
{  
  "primaryCif": "123456",  
  "familyName": "AL QAHTANI",  
  "dependents": [  
    {  
      "nationalId": "1234567890"  
    }  
  ]  
}
```

Response

```
{  
  "familyGroupId": "FG10001",  
  "status": "PENDING_OTP"  
}
```

24.2 Validate Family API

```
POST /api/v1/family/validate
```

Response Codes

Code	Meaning
200	Success
400	Invalid request
401	Unauthorized
409	Already linked
422	Validation failed
500	Internal error

24.3 Delink API

```
POST /api/v1/family/delink
```

Request

```
{
  "familyGroupId": "FG10001",
  "memberCif": "778899"
}
```

25. ERD / Data Model

25.1 Core Tables

FAMILY_GROUP

Column	Type	Description
FAMILY_GROUP_ID	PK	Unique identifier

Column	Type	Description
PRIMARY_CIF	FK	Main customer
FAMILY_NAME	VARCHAR	Family name
STATUS	VARCHAR	Current status
CREATED_DATE	DATETIME	Creation timestamp

FAMILY_MEMBER

Column	Type	Description
MEMBER_ID	PK	Member identifier
FAMILY_GROUP_ID	FK	Group reference
CIF	VARCHAR	Customer CIF
NATIONAL_ID	VARCHAR	Identity number
RELATIONSHIP_TYPE	VARCHAR	Spouse/Minor/Major
STATUS	VARCHAR	Link status

FAMILY_AUDIT

Column	Type	Description
AUDIT_ID	PK	Audit identifier
EVENT_TYPE	VARCHAR	Action type
USER_ID	VARCHAR	Initiator
CHANNEL	VARCHAR	Source channel
EVENT_DATE	DATETIME	Timestamp

25.2 Relationships

```

FAMILY_GROUP 1 --- * FAMILY_MEMBER
FAMILY_GROUP 1 --- * FAMILY_AUDIT
FAMILY_GROUP 1 --- * FAMILY_NOTIFICATION

```

26. State Machine Design

26.1 Family Lifecycle States

```
INITIATED
|
PENDING_OTP
|
VALIDATION_IN_PROGRESS
|
ACTIVE
|
DELINK_PENDING
|
DELINKED
```

26.2 Error States

- OTP_FAILED
- VALIDATION_FAILED
- EXPIRED
- REJECTED
- SYSTEM_ERROR

27. Error Handling Matrix

Scenario	System Action	User Message	Retry
OTP Expired	Reject OTP	OTP expired	Yes
Abshar Timeout	Retry integration	Service unavailable	Yes
Duplicate Member	Block request	Already linked	No
Invalid Relationship	Reject request	Validation failed	No
Notification Failure	Queue retry	No impact	Yes
Segment Failure	Rollback transaction	Processing failed	Yes

28. Deployment Architecture

28.1 Environment Topology

```
graph TD;
    A[Internet Users] --> B[Load Balancer];
    B --> C[API Gateway Cluster];
    C --> D[Application Cluster];
    D --> E[Database Cluster];
```

28.2 Deployment Environments

Environment	Purpose
DEV	Development
SIT	Integration Testing
UAT	Business Validation
PROD	Live Production
DR	Disaster Recovery

28.3 High Availability

- Active-active application nodes
- Database replication
- Auto failover
- Load balancing
- Health monitoring

29. Integration Design

29.1 Integration Types

Integration	Type
Abshar	Synchronous API
OTP Service	Synchronous API
Notification Service	Asynchronous Queue
CRM	REST API
RIB	REST API

29.2 Retry Strategy

Integration	Retry Count	Timeout
Abshar	3	30 sec
OTP	2	10 sec
Notification	5	Async

29.3 Queue Design

Queues

- FAMILY.NOTIFICATION
- FAMILY.AUDIT
- FAMILY.SEGMENT.UPDATE
- FAMILY.DELINK.EVENT

30. Security Design Deep Dive

30.1 Authentication

- OAuth2 authentication
- JWT token validation
- OTP verification

- Session timeout handling
-

30.2 Authorization Matrix

Role	Create	View	Delink	Approve
Primary Customer	Yes	Yes	Yes	No
Dependent	No	Limited	Self only	No
RM	Yes	Yes	Yes	No
Segmentation Unit	No	Yes	No	Yes

30.3 Data Protection

- AES-256 encryption at rest
 - TLS 1.2+ encryption in transit
 - Account masking
 - PII protection
 - Secure audit retention
-

30.4 Audit Retention

Audit Type	Retention
Customer Activities	10 years
OTP Logs	1 year
Notification Logs	3 years
Compliance Logs	10 years

31. Batch / EOD Design

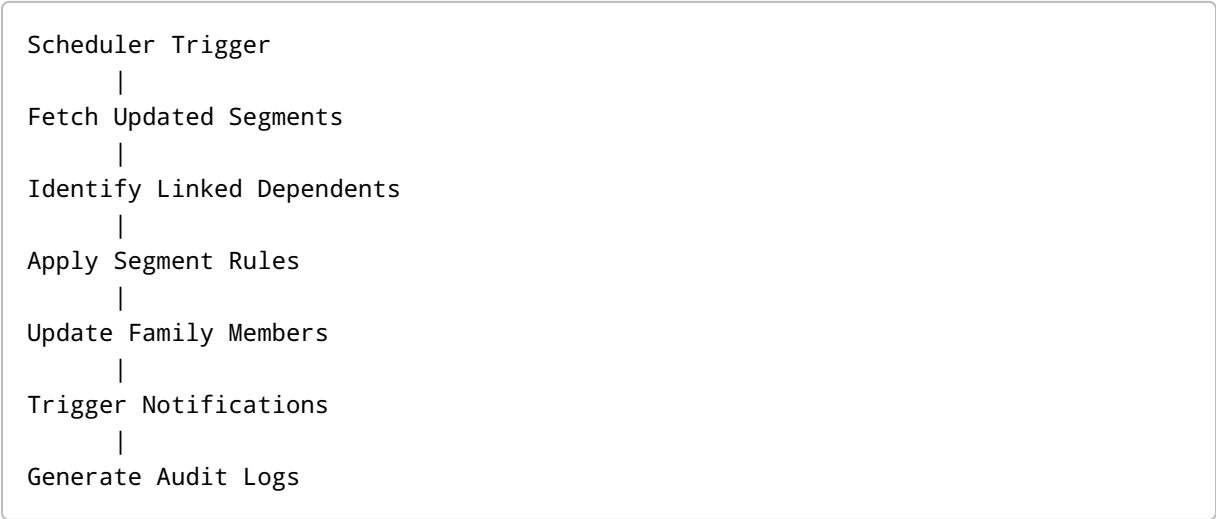
31.1 Batch Processing Scope

The batch engine will support:

- Segment upgrades
- Segment downgrades

- Family relationship reconciliation
- Notification retries
- Expired OTP cleanup
- Delink reconciliation

31.2 EOD Processing Flow



31.3 Batch Failure Handling

Scenario	Action
Batch Failure	Restart from checkpoint
Partial Failure	Retry failed records
Notification Failure	Move to retry queue
Data Mismatch	Raise reconciliation alert

31.4 Scheduler Frequency

Job	Frequency
Segment Propagation	Daily EOD
Notification Retry	Every 15 mins
OTP Cleanup	Hourly

Job	Frequency
Reconciliation	Daily

32. Conclusion

The Family Banking Solution establishes the foundational architecture for SAIB family relationship banking. The design supports scalable onboarding, secure validation, lifecycle management, segment inheritance, and customer engagement across CRM and digital banking channels while maintaining compliance, auditability, and operational scalability.